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Beyond Explicit Language: Plug-and-Play Visual-to-Linguistic Modeling Toward General Object Tracking

Kaiyang Lan[1,2†] Ying Cui[1,2†] Chenchen Jing[1,2] Jianwei Zheng[1,2] Dongyan Guo[1,2*]

[1] College of Computer Science and Technology, Zhejiang University of Technology

[2] Zhejiang Key Laboratory of Visual Information Intelligent Processing

Abstract

Natural language provides valuable auxiliary information for enhancing visual object tracking. While existing vision-language tracking methods explicitly leverage linguistic descriptions to aid tracking, they suffer from two critical limitations: the inability to dynamically adapt descriptions to the moving target and changing context; and the strong dependency on language input may cause failure when text is unavailable. To address the issues, we design a simple yet effective plug-and-play module that leverages linguistic assistance implicitly, without requiring explicit language input. The proposed textual inversion module converts visual features from template and search regions into text tokens in the CLIP text embedding space. It effectively inverts visual representations into linguistic forms, integrating contextual information from the both template and search regions. The linguistic cues are then injected into the visual feature space via a multi-layer semantic injection mechanism. The design enhances the completeness of cross-modal feature representations and the accuracy of inter-modal semantic alignment, thus enabling dynamically updated linguistic information guidance for general object tracking. Extensive experiments demonstrate the effectiveness of our proposed method. We integrate the proposed module into several advanced trackers and evaluate on both visual and vision-language tracking datasets, including MCITrack, DUTrack, and SeqTrack. By training only the newly introduced module and the corresponding decoder, the proposed approach achieves significant performance gains with minimal computational overhead. Code will be made publicly available.

1. Introduction

Visual object tracking is a fundamental task in computer vision. It aims to continuously predict the position of a target throughout a video sequence given only its initial state in the first frame. While traditional visual object trackers [2, 4, 15, 42, 44] rely solely on visual cues, they often struggle in challenging scenarios where visual information is insufficient or ambiguous. As illustrated in Fig. 1(a), tracking a specific sheep within a grouped herd is prone to failure due to high visual similarity among distractors. In such cases, additional textual guidance, such as a positional relationship description, can provide critical disambiguating information.

Natural language contains rich and high-level semantic

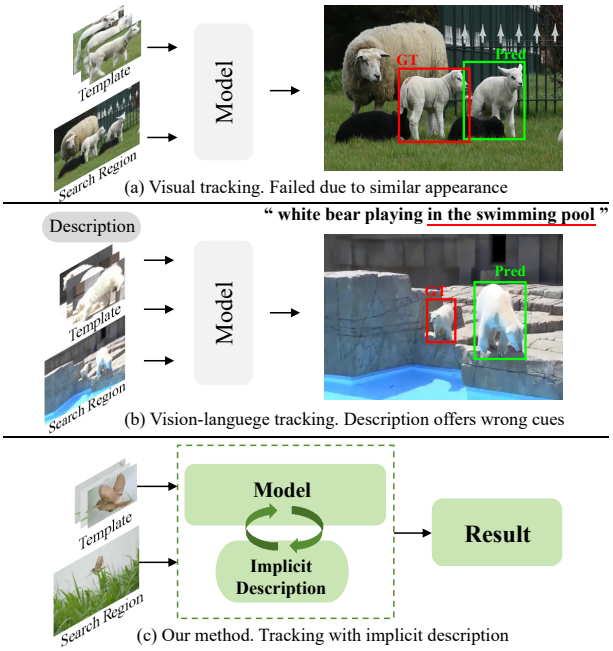


Figure 1. Comparison of different tracking pipelines. (a) A visual tracking pipeline relies on pure visual information on the left and a failure tracking example on the right. (b) A vision-language tracking pipeline relies on visual information and explicit natural language description on the left and an example of bias between the language description and the tracking object on the right. (c) Our proposed pipeline incorporates implicit description generation and semantic guidance for tracking.

†These authors contributed equally to this work.
*Corresponding author: guodongyan@zjut.edu.cn

information that can enhance visual object tracking by offering cues such as object attributes, appearance details, and contextual knowledge. Benefiting from the advances of natural language processing models, vision-language tracking (VLT) methods [13, 19, 30, 46, 48] have emerged by incorporating natural language descriptions as an additional input. However, existing VLT paradigms suffer from two principal limitations. First, they depend on pre-defined, static language descriptions, which often fail to align with the dynamically evolving context of a moving target. For example, as shown in Fig. 1(b), a description stating "the bear in the pool" becomes ineffective once the bear moves onto the rocks. Second, these methods exhibit a strong dependency on language input; if a textual description is unavailable, they typically regress to a vision-only model or fail entirely. While generating frame-specific descriptions online could resolve the inconsistency, it introduces significant practical overhead. That is, manual annotation is infeasible at scale, and leveraging large language models (LLMs) during inference severely compromises efficiency.

To overcome these challenges, we propose to design a novel plug-and-play module that dynamically generates and leverages implicit linguistic representations, thus eliminating the need for explicit text input. As depicted in Fig. 1(c), the proposed method is centered around a Textual Inversion Module. The module functions as an adaptor that converts visual embeddings from both the template and search regions into semantic tokens within the CLIP text embedding space. By processing the concatenated visual features, it effectively inverts them into a linguistic form and integrates the latent contextual information of the template into the semantic representation of the search region. Crucially, the inversion is performed at the patch level to yield fine-grained linguistic cues that reflect the current state of the target.

Subsequently, we design a Multi-Layer Semantic Injection Mechanism to fuse the linguistic representations back into the visual tracking pipeline. The visual and textual backbones are divided into corresponding blocks. For each block pair, a semantic-guided cross-attention module is employed to modulate the distribution of visual features using the inferred linguistic guidance. The multi-layer feature injection strategy ensures that shallow-level foundational visual features are effectively enriched with deep-level semantic information. Hence, it enhances the completeness of cross-modal representations and the accuracy of inter-modal alignment.

Extensive experiments demonstrate the effectiveness of our proposed method. We integrate our module into several state-of-the-art trackers and observe consistent performance improvements across multiple benchmarks. Notably, our method accomplishes it without any external language input and introduces only minimal computational overhead, as only the new module and the corresponding decoder require training.

In summary, the contributions of the work are as follows:

- We propose a novel Textual Inversion Module that dynamically generates consistent, implicit pseudo-descriptions from visual features to enhance tracking robustness. The module can be served as a plug-and-play one to improve the performance of existing trackers.
- We introduce a Multi-Layer Semantic Injection Mechanism that fuses linguistic information into visual features through a structured, multi-stage cross-attention process. The mechanism can effectively adapt the Textual Inversion Module into existing visual trackers.
- The proposed method demonstrates strong generalization ability. It improves the performance of various state-of-the-art trackers on both visual and vision-language benchmarks without requiring additional input modalities.

2. Related Work

2.1. Visual Tracking

Current object tracking methods can be broadly divided into image-level and video-level approaches. Image-level trackers rely solely on appearance matching between the template frame and search frame. Early Siamese-based two-stream methods [8, 17, 18] localize the target through similarity matching, while more recent one-stream methods [5, 9, 43] jointly encode the template and search regions, which simplifies the pipeline and achieves remarkable performance. Despite their efficiency, these methods often struggle with complex scenarios such as object deformation and motion blur due to their limited temporal reasoning.

Video-level trackers leverage video clips as input to integrate richer spatio-temporal context and achieve significant performance gains. VideoTrack [39] employs a video transformer backbone to blend spatiotemporal information. ODTrack [47] densely associates the contextual relationships in an online token propagation manner. Many video-level trackers adopt an auto-regressive approach. ARTrack [38], ARTrackV2 [1] jointly encode video clips and previous coordinate sequences to predict current coordinates in an auto-regressive manner. AQATrack [40] introduces auto-regressive queries to capture contextual information using a sliding window. Furthermore, State Space Models have also been explored for continuous contextual modeling, as seen in MCITrack [23].

While these video-level methods focus primarily on visual representations, our work extracts and leverages semantic information from image sequences to guide the tracking process.

2.2. Vision-language Tracking

Vision-language tracking is typically conducted in two ways: using only a natural language specification [33, 48],

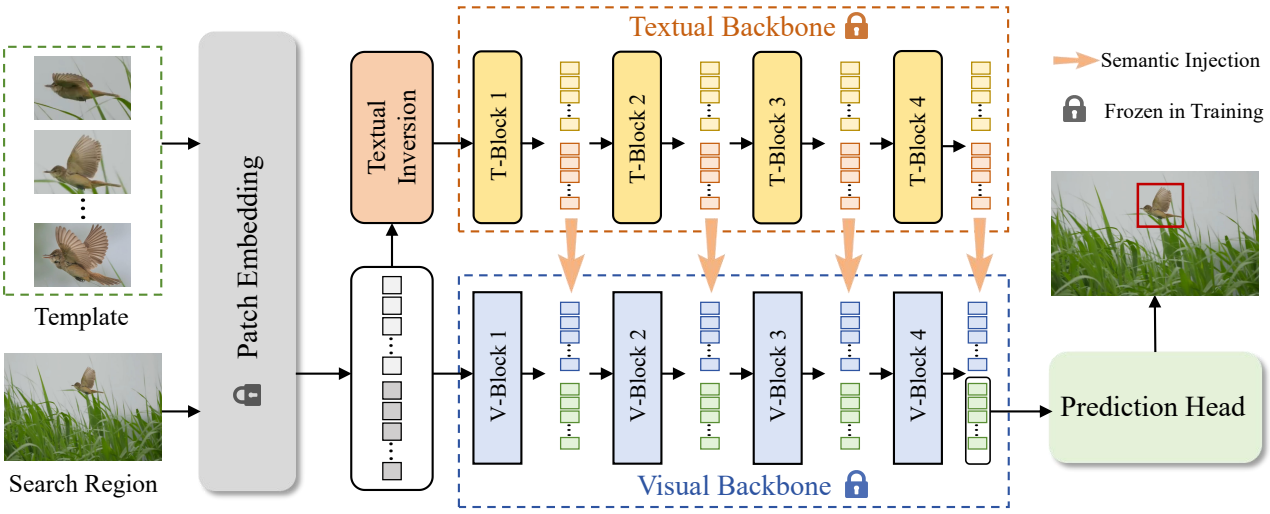


Figure 2. The overall framework of Trackers with the proposed TIMI. The input and output remain consistent with the original tracker. The Patch Embedding, Vision Backbone and the Prediction Head are the components of the original tracker.The key components in our method include the Textual Inversion Module, the Textual backbone for linguistic feature extraction, and Semantic Injection Module for integrating visual and linguistic features. The Textual Backbone is uniformly divided into N parts, which are referred to as S-Blocks, and the Vision Backbone is uniformly divided into N parts, which are referred to as V-Blocks.

or using both an initial bounding box and a language specification [26, 30, 37]. Li et al. [27] first define this task of tracking the target by lingual specification only. They employ separate grounding and tracking models to accomplish this task. JointNLT [48] simplifies the architecture with a unified Transformer-based model to handle both grounding and tracking. QueryNLT [33] introduces a filtering mechanism to address the modality inconsistencies. Tracking using both an initial bounding box and a language specification aims to improve tracking performance with language assistance [36]. SNLT [14] embeds a natural language description into a convolutional kernel to dynamically aggregate multimodal features, thereby enhancing the performance of some Siamese trackers. VLT [20] augments visual target representations by re-weighting visual features with language embeddings. UVLTrack [30] proposes a multimodal contrastive loss to align the visual and language features into a unified semantic space. DUTrack [26] addresses potential inconsistencies by dynamically updating both visual and language references.

These methods all rely on explicit natural language input, whereas our approach utilizes implicit linguistic information derived directly from visual features.

2.3. Linguistic Guidance in Tracking

The introduction of natural language into tracking [27] has shown that linguistic guidance can significantly improve performance. Although benchmarks like LaSOT [12], GOT-10K [22], and TNL2K [35] support vision-language tracking with their language annotations, these annotations are static. Such limitation leads to a critical problem of semantic drift and hampers the effectiveness of linguistic assistance. When the linguistic description differs from the actual state of the target, it may fail to provide useful guidance and can potentially mislead the model predictions. Several attempts have been made to address the above issue. DTVLT [25] employs a large language model to automatically generate textual annotations for representative tracking benchmarks at specific frame intervals, but inconsistencies still exist. QueryNLT [33] filters out inconsistent text, but this approach reduces the amount of valuable information. DUTrack [26] employs BLIP to dynamically generate descriptions during inference, which significantly slows down the inference speed. Moreover, the generated description differs in style from the annotations in training.

In contrast, our method generates implicit linguistic representations directly from fine-grained visual embeddings. It avoids the need for manual annotation or external captioning models, enabling efficient and consistent linguistic guidance throughout the tracking process.

3. Method

3.1. Preliminaries

Our method is designed for current mainstream transformer-based one-stream tracking frameworks. These trackers take the following information as input: a video clip containing N frames $\mathbf{V} \in \mathbb{R}^{N\times 3\times H_z\times W_z}$ or N

discrete template images $\mathbf{Z} \in \mathbb{R}^{N\times 3\times H_z\times W_z}$, along with a search image $\mathbf{X} \in \mathbb{R}^{3\times H_x\times W_x}$, and output the target bounding box in the search image. The main structure of these trackers can be divided into three components: first, a patch embedding layer that converts images into processable vectors; second, a visual backbone composed of stacked transformer layers, which progressively extracts visual features from shallow to deep levels; third, a prediction head that serves as a decoder to output the bounding box from visual features.

Vision-language trackers mainly build on the aforementioned trackers by adding a text encoder for extracting textual features from additional text inputs, and operations for fusing semantic and visual features.

3.2. Overview

The overall architecture of trackers with TIMI is shown in Fig. 2. Although a pure visual tracker is used for demonstration in the figure, our method can also equally well incorporated in vision-language trackers to enhance their feature representations. We divide the original tracker into three parts: patch embedding, vision transformer blocks, and decoder. The visual transformer blocks are formed by uniformly dividing the stacked transformer layers from the original backbone into four groups. We introduce TIMI to optimize feature extraction of the vision backbone. TIMI consists of two parts: Textual Inversion Module and Multi-Layer Semantic Injection Mechanism. It generates pseudo-descriptions for input images at the patch level and progressively guides the transformation of visual features through multi-layer feature injection from shallow to deep layers.

3.3. Textual Inversion Module

The Textual Inversion Module is designed to dynamically generate semantic representations that complement visual cues during tracking. It comprises two main components: a pseudo-description generator and a textual backbone. Together, they bridge the visual and linguistic domains, enabling the model to produce adaptive textual representations aligned with the state of the visual target.

Given the template and search regions, we first obtain their visual embeddings through a patch embedding layer. These embeddings are then concatenated to form a unified representation, which serves as the input to the pseudo-description generator. The generator transforms the concatenated embedding into the semantic space of the language backbone, effectively converting visual context into a text-aligned latent representation.

During this process, we model global contextual interactions between the template and search embeddings, allowing the network to integrate semantic cues of the target across both spatial and temporal dimensions. Specifically, the latent contextual information from the template is infused into the search embedding, enhancing its discriminative capability. Furthermore, the transformation is performed at the patch level, where each patch corresponds to a text token. This fine-grained mapping ensures that localized visual details are faithfully captured in the generated pseudo-description.

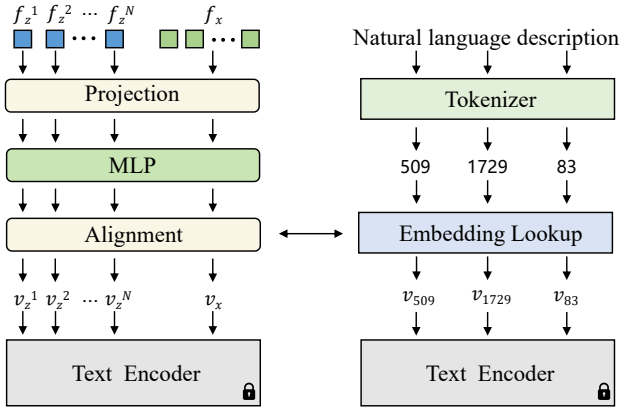


Figure 3. Outline of the Textual Inversion Module. The left part is our proposed Textual Inversion Module, which takes visual embeddings as input and outputs pseudo-descriptions. The right part illustrates the processing of natural language with the CLIP model.

As illustrated in Fig. 3, the Textual Inversion Module converts visual embeddings into a set of vectors to be fed into the text encoder through Projection, MLP, and Alignment operations. These vectors are referred to as pseudo-descriptions. Compared with the embedding process of standard natural language, the Textual Inversion Module can be regarded as functioning both as a Tokenizer and an Embedding Lookup layer.

The resulting pseudo-description embeddings are subsequently processed by a frozen CLIP text encoder to extract high-level linguistic features. These text features serve as dynamic, context-aware language cues that assist the visual backbone in tracking, providing consistent implicit semantic guidance.

3.4. Multi-Layer Semantic Injection Mechanism

The visual encoder consists of a patch embedding layer and transformer layers.It takes a template video clip $\mathbf{V}$ containing $\mathbf{N}$ template frames or N discrete template images $\mathbf{Z}$ and a search region $\mathbf{X}$ as input. First, both the template and search regions are processed through patch embedding layers, which consist of convolutional layers and MLP layers with down-sample ratio R into patches $\mathbf{P_v} \in \mathbb{R}^{N\times C\times \frac{H_z}{R}\times \frac{W_z}{R}}$ and $\mathbf{P_x} \in \mathbb{R}^{C\times \frac{H_x}{R}\times \frac{W_x}{R}}$. These patches are concatenated along the spatial dimension to form $\mathbf{P_{vx}} \in \mathbb{R}^{C\times (N\times \frac{H_z}{R}\times \frac{W_z}{R}+\frac{H_x}{R}\times \frac{W_x}{R})}$, which is then input into the transformer layers to extract visual features.

To adjust visual feature extraction from multiple lev-

els, we uniformly divide the transformer layers into four blocks named V-Blocks, corresponding to visual features $\mathbf{F}_{vx}^{i} \in \mathbb{R}^{C\times(N\times\frac{H_z}{R}\times\frac{W_z}{R}+\frac{H_x}{R}\times\frac{W_x}{R})}$ from shallow to deep, where $i \in \{1, 2, 3, 4\}$.

We also uniformly divide the transformer layers in CLIP text encoder into four groups named T-Block corresponding to text features $\mathbf{F}_{s}^{i}$ with the same shape as $\mathbf{F}_{vx}^{i}$, where $i \in \{1, 2, 3, 4\}$. These layers take pseudo-descriptions generated by the Textual Inversion Module as input.

For paired visual feature $\mathbf{F}_{vx}^{i}$ and text feature $\mathbf{F}_{s}^{i}$, we adjust $\mathbf{F}_{vx}^{i}$ with the guidance of $\mathbf{F}_{s}^{i}$ through the proposed Semantic Injection Module. First, $\mathbf{F}_{s}^{i}$ passes through an Alignment Module consisting of MLP layers. Second, we propose a Injection Module to adjust the visual features and obtain the updated $\mathbf{F}_{vx}^{i'}$, which serves as the input to the next block. The processing of the Multi-Layer Semantic Injection Mechanism can be summarized as follows:

$$\begin{aligned}
\mathbf{F}_{vx}^{i} &= \text{V-Block}^{i}(\mathbf{F}_{vx}^{i-1'}),\\
\mathbf{F}_{s}^{i} &= \text{T-Block}^{i}(\mathbf{F}_{s}^{i-1}),\\
\mathbf{F}_{s}^{i'} &= \text{Alignment}^{i}(\mathbf{F}_{s}^{i}),\\
\mathbf{F}_{vx}^{i'} &= \text{Injection}^{i}(\mathbf{F}_{s}^{i'}, \mathbf{F}_{vx}^{i}),
\end{aligned} \tag{1}$$

where V-Block^i, T-Block^i, Alignment^i and Injection^i represent the visual extraction block, textual extraction block, Alignment module and Injection Module, respectively, in the i-th integration.

In the Injection Module, we normalize $\mathbf{F}_{s}^{i}$ and $\mathbf{F}_{vx}^{i}$ separately. Then, $\mathbf{F}_{s}^{i}$ is used as Query Q, while $\mathbf{F}_{vx}^{i}$ is used as Key K and Value V, in the Multi-Head Attention layer. Subsequently, the cross-attention results are multiplied by a learnable scaling factor α^i and then added to $\mathbf{F}_{vx}^{i}$ to obtain the adjusted $\mathbf{F}_{vx}^{i'}$. The processing of the Semantic Injection module can be summarized as follows:

$$\begin{aligned}
\mathbf{Q}^{i} &= \text{Norm}^{i}(\mathbf{F}_{vx}^{i}),\\
\mathbf{K}^{i}, \mathbf{V}^{i} &= \text{Norm}^{i}(\mathbf{F}_{s}^{i'}),\\
\mathbf{Attn}^{i} &= \text{MHCA}^{i}(\mathbf{Q}^{i}, \mathbf{K}^{i}, \mathbf{V}^{i}),\\
\mathbf{F}_{vx}^{i'} &= \mathbf{F}_{vx}^{i} + \alpha^{i} \times \mathbf{Attn}^{i},
\end{aligned} \tag{2}$$

where Norm^i, α^i, MHCA^i and Attn^i represent the LayerNorm, learnable scaling factor, Multi-Head Cross-Attention calculation and its resulting attention output, respectively, in the i_{th} Injection Module. In this way, the visual features are adjusted with the semantic guidance.

4. Experiments

4.1. Experiment Settings

Model. To validate the effectiveness of our method, we select several baselines: the state-of-the-art model in Vision-Only Tracking (MCITrack [23]), the state-of-the-art model in Vision-Language Tracking (DUTrack [26]), and an earlier influential model in Vision-Only Tracking (SeqTrack [7]). MCITrack and DUTrack use Fast-iTPN [34] as visual backbone. Fast-iTPN consists of transformer layers similar to ViT. SeqTrack uses a standard ViT [11] as visual backbone.

Table 1. Model parameters, FLOPs, and inference speed. In the evaluation of DUTrack, BLIP [24] as the text generator during the inference phase was not included.

Model variants	Params(M)	FLOPS(G)	Speed(fps)
SeqTrack-B256	89	66	40
SeqTrack-B256*	129	98	29
MCITrack-B224	88	38	31
MCITrack-B224*	131	58	20
DUTrack-B256	70	35	44
DUTrack-B256*	108	53	33

Additionally, the pre-trained CLIP [32] text model used in our method is clip-vit-large-patch14, with 12 transformer layers in the backbone. It has approximately 40G FLOPs and 85M parameters.

Notably, the implementation of our method results in a FPS decrease of roughly 10, of which 7 FPS comes from the CLIP text encoder. Our proposed Textual Inversion Module and Semantic Injection Module contribute marginally to the overall speed loss. Furthermore, the visual backbone Fast-iTPN, adopted in MCITrack and DUTrack is deeper yet narrower compared to the standard ViT employed in SeqTrack. Consequently, despite having lower computational complexity, MCITrack and DUTrack exhibit slower inference speeds.

Training and Inference. Our training process consists of two stages. In the first stage, we follow the settings of the original trackers, enabling the visual backbone to extract high-quality features.

In the second stage, we integrate our proposed modules into the trained trackers while keeping both the visual and textual backbones frozen. Only the Textual Inversion Module, Semantic Injection, and Decoder are trained. This strategy focuses learning on mapping visual features to the frozen textual space and leveraging visual-linguistic priors from the pre-trained CLIP model. All models are trained for 10 epochs, with the learning rate scheduled to decay after the 6th epoch.

4.2. Comparisons of Experimental Results

During the model comparison experiments, we follow the same configurations and hyperparameters as reported in the original work, to better validate the effectiveness of our proposed method.

LaSOT is a large-scale long-term tracking benchmark that includes 1,120 sequences for training, 280 sequences

Table 2. Performance comparison on four large-scale benchmarks, including LaSOT [12], GOT-10K [22], TrackingNet [31], and TNL2K [35]. These state-of-the-art methods are divided into vision-only methods and vision-language methods. We add a symbol * to a method to indicate that the corresponding model is integrated with our proposed method. Better results are highlighted with underlined fonts, and the top results are highlighted with **bold**.

Type	Method	Source	LaSOT			GOT-10K			TrackingNet			TNL2K		
			AUC	P_{Norm}	P	AO	$SR_{0.5}$	$SR_{0.75}$	AUC	P_{Norm}	P	AUC	P_{Norm}	P
Vision-only	STARK [41]	ICCV2021	67.1	77.0	-	68.8	78.1	64.1	82.0	86.9	-	-	-	-
	TransT [6]	CVPR2021	64.9	73.8	69.0	67.1	76.8	60.9	81.4	86.7	80.3	50.7	-	-
	OSTrack-B256 [43]	ECCV2022	69.1	78.7	75.2	71.0	80.4	68.2	83.1	87.8	82.0	54.3	-	-
	ARTrack-B256 [38]	CVPR2023	70.4	79.5	76.6	73.5	82.2	70.9	84.2	88.7	83.5	57.5	-	-
	AQATrack-B256 [40]	CVPR2024	71.4	81.9	78.6	73.8	83.2	82.1	83.8	88.6	83.1	57.8	-	59.4
	ARTrackV2-B256 [1]	CVPR2024	71.6	80.2	77.2	75.9	85.4	72.7	84.9	89.3	84.5	59.2	-	-
	ODTrack-B384 [47]	AAAI2024	73.2	83.2	80.6	77.0	87.9	75.1	84.8	89.4	83.7	60.9	-	-
	ARPTrack-B256 [28]	CVPR2025	72.6	81.4	78.5	77.7	87.3	74.3	85.5	90.0	85.3	-	-	-
	DreamTrack-B256 [21]	CVPR2025	73.8	83.4	80.6	77.5	87.1	74.2	85.8	90.0	85.3	60.4	-	63.2
	SPMTrack-B378 [3]	CVPR2025	74.9	84.0	81.7	76.5	85.9	76.3	86.1	90.2	85.6	62.0	79.7	66.7
	SeqTrack-B256 [7]	CVPR2023	69.9	79.7	76.3	74.7	84.7	71.8	83.3	88.3	82.2	54.9	-	-
	SeqTrack-B256*	Ours	71.0	81.0	77.4	76.6	85.7	74.5	83.4	88.4	82.4	56.5	74.1	58.1
	MCITrack-B224 [23]	AAAI2025	75.3	85.6	83.3	77.9	88.2	76.8	**86.3**	**90.9**	**86.1**	62.9	-	-
	MCITrack-B224*	Ours	**76.1**	**86.4**	**84.4**	**79.4**	**89.2**	**80.0**	**86.3**	**90.9**	**86.1**	**63.1**	-	-
Vision-Language	JointNLT [48]	CVPR2023	60.4	69.4	63.6	-	-	-	-	-	-	56.9	-	58.1
	MMTrack-B384 [45]	TCSVT2023	70.0	82.3	75.7	-	-	-	-	-	-	58.6	75.2	59.4
	ATTrack [16]	MM2024	63.7	-	67.3	-	-	-	-	-	-	56.9	75.0	64.7
	UVLTrack-B256 [30]	AAAI2024	69.4	-	75.9	49.2	-	55.8	62.7	-	65.4	63.1	-	66.7
	QueryNLT [33]	CVPR2024	59.9	69.6	63.5	-	-	-	-	-	-	57.8	58.7	75.6
	MambaVLT [29]	CVPR2025	66.6	77.3	71.0	-	-	-	-	-	-	66.5	90.9	69.9
	DUTrack-B256 [26]	CVPR2025	73.0	83.8	81.1	76.7	-	-	-	-	-	64.9	82.9	70.6
	DUTrack-B256*	Ours	**73.6**	**84.3**	**81.6**	**78.3**	**89.3**	**77.3**	-	-	-	**67.2**	**85.6**	**73.0**

for testing, with initial language annotations. As reported in Tab. 2, our method improves the tracking performance. SeqTrack-B256* outperforms the original SeqTrack-B256, with improvements of 1.5%, 1.5% and 1.5%, in AUC, P_{Norm} and P, respectively. MCITrack-B224* outperforms the original MCITrack-B224 by 1.1%, 0.9%, 1.3% on these three metrics. DUTrack-B256* outperforms the original DUTrack-B256 by 0.9%, 0.5%, 0.7% on these three metrics.

TrackingNet is a comprehensive dataset containing 511 videos, which covers diverse object categories and scenes. As presented in Tab. 2, our method slightly improves the tracking performance. SeqTrack-B256* performs better than the original SeqTrack-B256, getting improvements of 0.1%, 0.2%, and 0.2% in AUC, P_{Norm}, and P, respectively. The two sets of results are highly comparable. Furthermore, MCITrack-B224* achieves the same performance as the base model.

Compared with LaSOT and GOT-10k, our model shows smaller gains on TrackingNet. This can be attributed to the characteristics of the dataset: TrackingNet consists of large-scale, in-the-wild videos collected from the Internet, which contain higher annotation noise, severe occlusions, and more dynamic camera motion. Such conditions reduce the reliability of high-level semantic information extracted from visual features and weaken the effectiveness of semantic-guided cues, making the tracker rely more on low-level visual robustness. In contrast, LaSOT and GOT-10k provide cleaner visual data and more consistent target appearances, where our model can fully exploit the integration of visual and linguistic information. Therefore, the relatively modest improvement on TrackingNet reflects the challenge of applying semantic-based tracking in highly unconstrained, real-world scenarios.

GOT-10K is a large-scale tracking dataset that contains more than 10,000 video sequences. The benchmark uses two metrics: Average Overlap (AO) and Success Rate (SR). Following the official protocol, we only use the GOT-10K training set to train our models. As reported in Tab. 2, SeqTrack-B256* performs better than the original SeqTrack-B256, with improvements of 2.5%, 1.2%, and 3.8% in AO, $SR_{0.5}$, and $SR_{0.75}$, respectively. MCITrack* outperforms the original MCITrack by 2.0%, 1.1%, and 4.2% on these three metrics. DUTrack-B256* outperforms the original DUTrack-B256 by 2.1% in Average Overlap.

TNL2K is a large-scale dataset with 700 challenging video sequences and initial language annotations. As shown in Tab. 2, DUTrack-B256* achieves 67.2%, 85.6%, and 73.0% in terms of AUC, P_{Norm}, and P, respectively. It outperforms the original DUTrack-B256 by 3.5%, 3.2% and 3.3% on these three metrics. SeqTrack-B256* achieves 56.5%, 74.1%, and 58.1% on three metrics. It outperforms the original SeqTrack by 1.6% on AUC metric.

4.3. Ablation study

In the ablation study, we investigate the impact of the pseudo-description, the Textual Inversion Module and the Multi-Layer Semantic Injection Mechanism on the model.

4.3.1. Study on the Pseudo-description

To verify the effectiveness of the pseudo-description, we conduct the experiments based on DUTrack-B256, which takes natural language as linguistic information assistance. The variable in this set of experiments is the source of semantic information. "Without description" indicates that we remove the natural language description of DUTrack-B256, converting it to pure visual tracking without linguistic assistance. "Pseudo-description only" indicates that we remove the natural language description of the baseline and use pseudo-description only to provide linguistic information. "Text and pseudo-description" indicates that we integrate our module into the baseline, using both natural language description and pseudo-description to to provide linguistic assistance to the model. As shown in Tab. 3, baseline achieves AUC 73.0%, P_{Norm} 83.8%, and P 81.1%. "Without description" results in scores of AUC 72.3%, P_{Norm} 83.4%, and P 80.3%. The decrease of this setting demonstrates the impact of language assistance. "Pseudo-description only" results in a 0.4% improvement over the baseline in AUC on LaSOT, and a 1.12% improvement over "Without description", highlighting the effectiveness of the pseudo-description. "Text and pseudo-description" results in a 0.6% improvement in AUC on LaSOT. This suggests that our method is applicable to vision-language tracker.

Table 3. Ablation study of the pseudo-description and Textual Inversion Module on LaSOT with DUTrack-B256 as the baseline. Δ denotes the performance change relative to the baseline.

Model variants	AUC	P_{Norm}	P	Δ
Baseline: text-only	73.0	83.8	81.1	-
Without description	72.3	83.4	80.3	-0.7
Pseudo-description only	73.4	84.2	81.4	0.4
Text and pseudo-description	73.6	84.3	81.6	0.6

4.3.2. Study on the Textual Inversion Module

To investigate the impact of textual backbone, we conduct experiments based on SeqTrack-B256*, which employs CLIP text encoder as the textual backbone. The variable in this set of experiments is the textual backbone. "None" indicates the absence of textual backbone. The pseudo-description generated by the Textual Inversion Module is copied and fed into the Semantic Injection Module directly, without passing through any text encoder. "With BERT" indicates that we replace the CLIP text encoder with BERT [10] to encode the pseudo-description into linguistic features. As shown in Tab. 4, using CLIP as textual backbone results in the scores of AUC 71.0%, P_{Norm} 80.9%, and P 77.4%. "None" results in a 0.5% decrease in AUC on LaSOT, demonstrating the importance of textual backbone for the Textual Inversion Module. "With BERT" results in a 0.6% decrease in AUC on LaSOT, achieving a performance close to that of "None". Such results may be attributed to the use of CLIP as a teacher model for the visual backbone during the pre-training phase. This suggests that textual backbone and visual backbone function most effectively when aligned through pre-training. Our method mines text-visual priors without requiring text input and effectively leverages them for tracking.

Table 4. Ablation study of the CLIP text encoder on the LaSOT dataset. Δ denotes the change in AUC relative to the baseline.

Model variants	AUC	P_{Norm}	P	Δ
Baseline: with CLIP	71.0	80.9	77.4	-
None	70.5	80.5	77.1	-0.5
With BERT	70.4	80.5	77.1	-0.6

4.3.3. Study on the Injection Mechanism

To investigate the impact of the Multi-layer Semantic Injection Mechanism, we adopt SeqTrack-B256* as the baseline, which divides the vision backbone and text backbone into 4 corresponding blocks. We set different integration times to investigate its impact on model performance and efficiency. "1 layer" denotes that the feature is injected after the whole backbone only with 1 layer. N layers denotes that both vision and text backbones are divided into N corresponding blocks, and injection are conducted after each block pair.

Table 5. Ablation study of the Multi-Layer Integration Mechanism on the LaSOT dataset with SeqTrack-B256* adopted as the baseline. Δ denotes the change in AUC relative to the baseline

Model variants	AUC	P_{Norm}	P	Δ	fps
baseline	71.0	80.9	77.4	-	29
1 layer	70.4	80.6	77.6	-0.6	36
2 layers	70.6	80.7	77.0	-0.4	34
3 layers	70.7	81.0	77.6	-0.3	31
6 layers	70.6	80.7	77.0	-0.4	26

As shown in Tab. 5, the "1 layer" setting results in the worst performance of 70.4% in AUC on LaSOT. The "2 layers" achieves an AUC of 70.6%, and the "3 layers" setting achieves an AUC of 70.7%. The "6 layers" setting achieves an AUC of 70.6%, which is close to the results of the "3 layers" setting. Besides, with the increase of interaction layers, the model becomes more complex and slower.

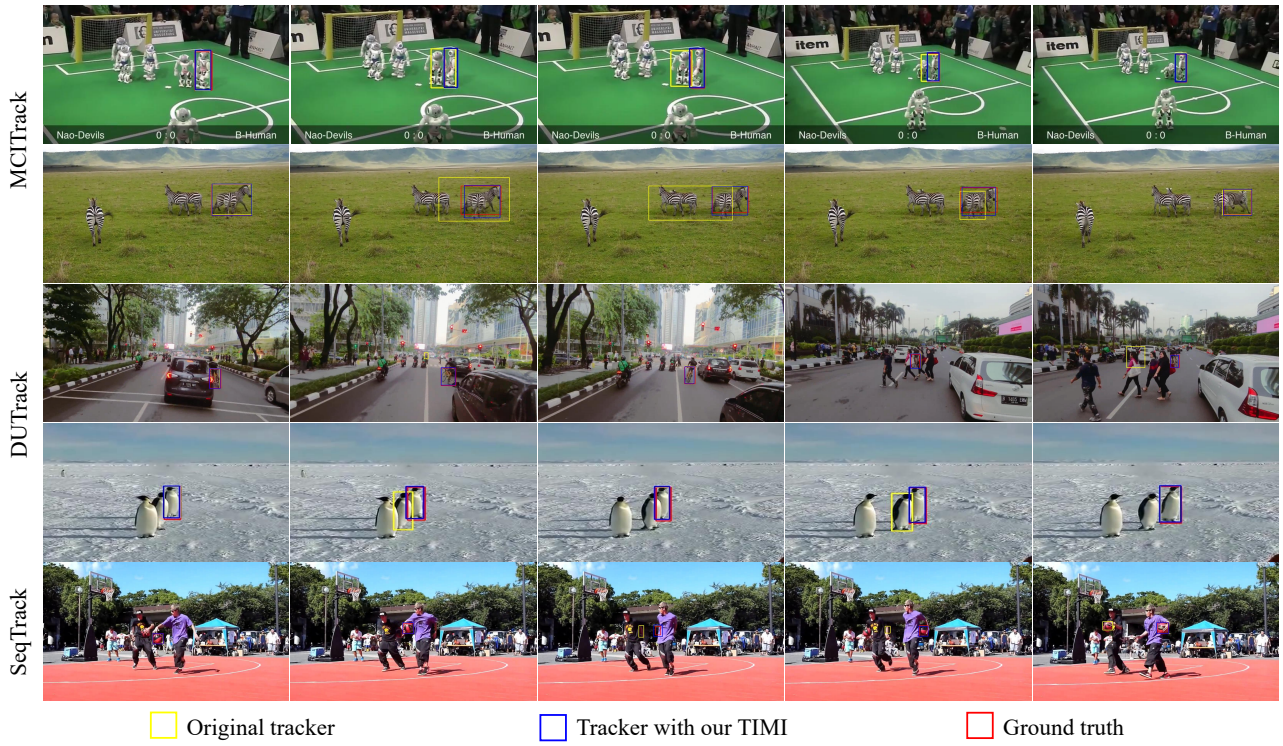


Figure 4. Qualitative comparison between trackers with our method and the original trackers on five challenging sequences from the LaSOT benchmark. The first two sequences present the visualization results of MCITrack and MCITrack*. The third and fourth sequences show the visualization results of DUTrack and DUTrack*. The last sequence displays the visualization results of SeqTrack and SeqTrack*.

The results demonstrates the effectiveness of the Multi-layer Semantic Injection Mechanism in adjusting the distribution of the visual features, while excessive interactions increase model complexity and may degrade performance.

4.4. Visualization

A visual comparison is shown in Fig. 4 with five challenging sequences shown the effectiveness of the proposed methods.

The first sequence shows tracking under challenges of similar appearances and partial occlusion. MCITrack fails and locks onto a similar object, while MCITrack* maintains correct tracking. The second sequence demonstrates that when confronted with the challenge of similar appearances, MCITrack tracks a similar object, whereas MCITrack* achieves accurate predictions. The third sequence illustrates that in the case of full occlusion, DUTrack cannot recover tracking when the target reappears, whereas DUTrack* restores tracking immediately. The fourth sequence reveals that when encountering the challenges of similar appearances and partial occlusion, DUTrack tracks a similar target, while DUTrack* achieves correct tracking. The fifth sequence indicates that under conditions of fast motion and similar appearances, SeqTrack tracks a similar target and fails to recover, whereas SeqTrack* sustains accurate tracking throughout.

From the qualitative comparison, it can be observed that our method provides effective assistance to the tracker when addressing these challenges. Compared to DUTrack that generates explicit natural language descriptions for each search frame, our method achieves more stable linguistic and semantic guidance.

5. Conclusion

In this paper, we address two key limitations in vision-language tracking: the reliance on static, pre-defined language descriptions and the performance degradation in the absence of textual input. To this end, we propose a novel plug-and-play module that dynamically generates implicit linguistic representations directly from visual features, eliminating the dependency on explicit language. Our approach centers on two simple components: a textual inversion module that converts visual embeddings into semantic tokens, and a multi-step integration mechanism that fuses these linguistic cues into the visual backbone. Extensive experiments demonstrate that the design significantly enhances the performance of various state-of-the-art trackers across multiple benchmarks, without requiring any additional language input and with minimal computational overhead.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant (No.62572438, No. 62506338), in part by the Zhejiang Provincial Natural Science Foundation of China under Grant (No.LZ25F020008, No. LMS25F020004).

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